

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 23 — Chemical Energetics

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Lattice Energy · Born–Haber Cycles · Hydration & Solution · Entropy S · Gibbs Free Energy G
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**23.1 Lattice Energy & Born–Haber Cycles****23.2 Enthalpies of Solution & Hydration****23.3 Entropy Change, S****23.4 Gibbs Free Energy Change, G****High-Frequency Core Repeats (Q41–Q50)**

- Standard lattice energy definitions; Born–Haber cycles for binary and ternary ionic solids; theoretical maximum lattice energy
- Definitions of H°_{sol} and H°_{hyd} ; Hess's Law cycles linking solution, hydration and lattice energy; theoretical maximum hydration enthalpy
- Concept of entropy and disorder; calculating S°_{sys} from standard molar entropies; S_{surr} and S_{tot}
- $G = H - TS$; feasibility conditions; calculation of feasibility temperatures; temperature dependence of G
- The 10 most frequently tested Cambridge Paper 4 questions on Chemical Energetics from the past 10 years

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Zinc Sulfide Born–Haber Cycle — 9701/42/M/J/23/Q1 [6 Marks]

HARD · Syllabus 23.1

Zinc sulfide, ZnS, occurs naturally as sphalerite. A Born–Haber cycle for ZnS(s) is shown in Fig. 1.1. Thermochemical data (kJ mol^{-1}): $\text{H}^\circ_f[\text{ZnS}] = -206$; $\text{H}^\circ_{\text{at}}[\text{Zn}] = +131$; $(\text{IE}_1 + \text{IE}_2)[\text{Zn}] = +2640$; $\text{H}^\circ_{\text{at}}[\text{S}] = +279$; $(\text{EA}_1 + \text{EA}_2)[\text{S}] = +332$.

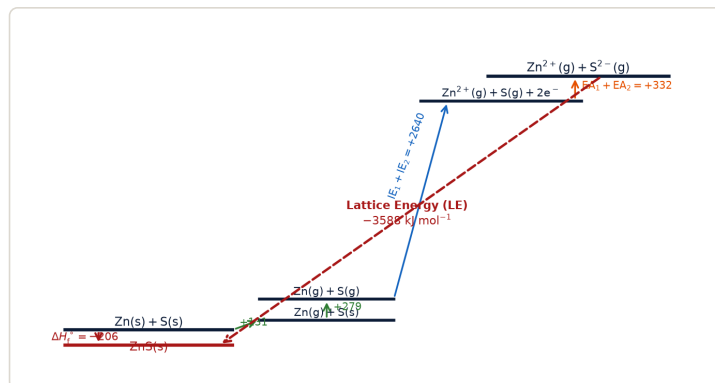


Fig. 1.1. Born–Haber cycle energy level diagram for crystalline zinc sulfide, ZnS(s).

- (a) Define the standard enthalpy change of hydration of an ion, $\text{H}^\circ_{\text{hyd}}$. [2]

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- (b) Construct the Born–Haber cycle expression and calculate the lattice energy of ZnS(s). [4]

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2 Magnesium Chloride Lattice Energy & Feasibility — 9701/41/M/J/23/Q2 [6 Marks]

HARD · Syllabus 23.1

Magnesium chloride decomposes: $\text{MgCl}_2(\text{s}) \rightarrow \text{Mg}(\text{s}) + \text{Cl}_2(\text{g})$, $\text{H}^\circ = +641 \text{ kJ mol}^{-1}$.
Data: $\text{H}^\circ_{\text{at}}[\text{Mg}] = +147$, $(\text{IE}_1 + \text{IE}_2)[\text{Mg}] = +2187$, $\text{BE}(\text{Cl}_2) = +242$, $\text{EA}(\text{Cl}) = -349 \text{ kJ mol}^{-1}$.
 S° values ($\text{J K}^{-1} \text{ mol}^{-1}$): $\text{MgCl}_2(\text{s}) = 90.0$; $\text{Mg}(\text{s}) = 32.7$; $\text{Cl}_2(\text{g}) = 223.0$.

- (a) Calculate the lattice energy of $\text{MgCl}_2(\text{s})$. [3]

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- (b) Calculate S° for the decomposition and deduce the minimum temperature at which it becomes feasible. [3]

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3 Calcium Fluoride Cycle & Solubility Thermodynamics — 9701/43/M/J/23/Q1 [6 Marks]

HARD · Syllabus 23.1

Data for CaF_2 (kJ mol^{-1}): $\text{H}^\circ_f = -1220$; $\text{H}^\circ_{\text{at}}[\text{Ca}] = +178$; $(\text{IE}_1 + \text{IE}_2)[\text{Ca}] = +1735$; $\text{BE}(\text{F–F}) = +158$; $\text{EA}[\text{F}] = -328$.
 $\text{H}^\circ_{\text{hyd}}(\text{Ca}^{2+}) = -1577$; $\text{H}^\circ_{\text{hyd}}(\text{F}^-) = -506$.

(a) Calculate the lattice energy of calcium fluoride, $\text{CaF}_2(\text{s})$. [3]

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(b) Calculate $\Delta H^\circ_{\text{sol}}$ of CaF_2 and explain why it is much less soluble than CaCl_2 . [3]

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4 Barium Sulfate Solution Cycle & Group 2 Trends — 9701/42/O/N/22/Q1 [6 Marks]

HARD · Syllabus 23.2

Lattice energy of $\text{BaSO}_4 = -2469 \text{ kJ mol}^{-1}$.

Enthalpies of hydration: $\Delta H^\circ_{\text{hyd}}(\text{Ba}^{2+}) = -1305 \text{ kJ mol}^{-1}$; $\Delta H^\circ_{\text{hyd}}(\text{SO}_4^{2-}) = -1145 \text{ kJ mol}^{-1}$.

(a) Construct an energy cycle and calculate the standard enthalpy of solution, $\Delta H^\circ_{\text{sol}}$, of BaSO_4 . [3]

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(b) Explain why the solubility of Group 2 sulfates decreases down the group from MgSO_4 to BaSO_4 . [3]

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5 Silver Chloride Lattice Energy: Experimental vs Theoretical — 9701/42/M/J/22/Q1 [6 Marks]

HARD · Syllabus 23.1

Data for AgCl (kJ mol^{-1}): $\Delta H^\circ_{\text{f}} = -127$; $\Delta H^\circ_{\text{at}}[\text{Ag}] = +285$; $\text{IEI}[\text{Ag}] = +731$; $\Delta H^\circ_{\text{at}}[\text{Cl}] = +121$; $\text{EA}[\text{Cl}] = -349$.

Theoretical electrostatic lattice energy = -770 kJ mol^{-1} .

(a) Calculate the experimental lattice energy of $\text{AgCl}(\text{s})$ using the Born–Haber cycle. [3]

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(b) Explain the discrepancy between the experimental and theoretical lattice energy values. [3]

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6 Copper(I) Oxide Born–Haber Cycle & Electron Affinities — 9701/42/O/N/21/Q1 [6 Marks]

HARD · Syllabus 23.1

Data for Cu_2O (kJ mol^{-1}): $\Delta H^\circ_{\text{f}} = -169$; $\Delta H^\circ_{\text{at}}[\text{Cu}] = +338$; $\text{IEI}[\text{Cu}] = +746$; $\Delta H^\circ_{\text{at}}[\text{O}] = +249$; $\text{EA1}[\text{O}] = -141$; $\text{EA2}[\text{O}] = +798$.

(a) Calculate the lattice energy of $\text{Cu}_2\text{O}(\text{s})$. [4]

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(b) Explain why EA1 of oxygen is exothermic whereas EA2 is endothermic. [2]

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7 Thermodynamics of Ammonium Nitrate Dissolution — 9701/41/O/N/21/Q2 [6 Marks]

HARD · Syllabus 23.2

$\text{NH}_4\text{NO}_3(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq})$, $H^\circ_{\text{sol}} = +25.7 \text{ kJ mol}^{-1}$.
 S° values ($\text{J K}^{-1} \text{ mol}^{-1}$): solid = 151; $\text{NH}_4^+(\text{aq}) = 113$; $\text{NO}_3^-(\text{aq}) = 146$.

(a) Calculate the standard entropy change of solution, S°_{sol} , at 298 K. [3]

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(b) Calculate G°_{sol} at 298 K and explain why dissolution occurs spontaneously despite being endothermic. [3]

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8 Thermal Decomposition of Calcium Carbonate — 9701/41/M/J/21/Q1 [6 Marks]

HARD · Syllabus 23.4

$\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$.
 H°_f (kJ mol^{-1}): $\text{CaCO}_3 = -1207$; $\text{CaO} = -635$; $\text{CO}_2 = -394$.
 S° ($\text{J K}^{-1} \text{ mol}^{-1}$): $\text{CaCO}_3 = 92.9$; $\text{CaO} = 39.7$; $\text{CO}_2 = 213.6$.

(a) Calculate standard enthalpy change H° and standard entropy change S° . [3]

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(b) Calculate the minimum temperature, in $^\circ\text{C}$, for decomposition to become feasible. [3]

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9 Polarization & Lattice Energy of Lithium Iodide — 9701/43/M/J/21/Q2 [6 Marks]

HARD · Syllabus 23.1

Data for LiI (kJ mol^{-1}): $H^\circ_f = -270$; $H^\circ_{\text{at}}[\text{Li}] = +161$; $\text{IEI}[\text{Li}] = +520$; $H^\circ_{\text{at}}[\text{I}] = +107$; $\text{EA}[\text{I}] = -295$.
 Theoretical electrostatic LE = -738 kJ mol^{-1} .

- (a) Calculate the experimental lattice energy of LiI(s) . [3]

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- (b) Explain why the experimental LE is more exothermic than theoretical LE in terms of ion polarization. [3]

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10 Complete Born–Haber Cycle for Aluminium Oxide — 9701/42/M/J/18/Q1 [6 Marks]

HARD · Syllabus 23.1

Al_2O_3 data (kJ mol^{-1}): $\text{H}^\circ_f = -1676$; $\text{H}^\circ_{\text{at}}[\text{Al}] = +326$; $(\text{IE1}+\text{IE2}+\text{IE3})[\text{Al}] = +5140$; $\text{BE}(\text{O}_2) = +496$; $(\text{EA1}+\text{EA2})[\text{O}] = +657$.

- (a) Calculate the lattice energy of aluminium oxide, $\text{Al}_2\text{O}_3(\text{s})$. [4]

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- (b) Explain why LE of Al_2O_3 is vastly more exothermic than that of Na_2O ($\text{LE} = -2478 \text{ kJ mol}^{-1}$). [2]

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11 Haber Process Synthesis Feasibility & Temperature Dependence — 9701/42/O/N/23/Q2 [6 Marks]

HARD · Syllabus 23.4

Ammonia synthesis: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\text{H}^\circ = -92.2 \text{ kJ mol}^{-1}$.

Standard entropies ($\text{J K}^{-1} \text{ mol}^{-1}$): $\text{N}_2 = 191.6$; $\text{H}_2 = 130.6$; $\text{NH}_3 = 192.3$.

- (a) Calculate the standard entropy change S° for this reaction. [2]

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- (b) Calculate G° at 298 K and deduce whether the reaction is spontaneous at room temperature. [2]

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- (c) Calculate the temperature at which the reaction ceases to be spontaneous ($\text{G}^\circ = 0$). [2]

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12 Potassium Chloride Solution Thermodynamics — 9701/41/O/N/23/Q1 [6 Marks]

HARD · Syllabus 23.2

KCl dissolves endothermically: $\text{KCl(s)} \rightarrow \text{K}^+(\text{aq}) + \text{Cl}^-(\text{aq})$.
Data: $\text{LE}(\text{KCl}) = -711 \text{ kJ mol}^{-1}$; $\text{H}^\circ_{\text{hyd}}(\text{K}^+) = -322 \text{ kJ mol}^{-1}$; $\text{H}^\circ_{\text{hyd}}(\text{Cl}^-) = -364 \text{ kJ mol}^{-1}$.
 S° : $\text{KCl(s)} = 82.6$; $\text{K}^+(\text{aq}) = 102.5$; $\text{Cl}^-(\text{aq}) = 56.5 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a)** Calculate the standard enthalpy of solution, $\text{H}^\circ_{\text{sol}}$, of KCl. **[2]**

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- (b)** Calculate the standard entropy change of solution, S°_{sol} . **[2]**

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- (c)** Calculate G°_{sol} at 298 K and explain why KCl dissolves readily despite $\text{H}^\circ_{\text{sol}}$ being endothermic. **[2]**

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13 Nickel(II) Oxide Born–Haber Cycle & Ligand Energy — 9701/43/O/N/23/Q1 [6 Marks]

HARD · Syllabus 23.1

Nickel(II) oxide, NiO, has rock-salt lattice.
Data (kJ mol^{-1}): $\text{H}^\circ_{\text{f}}[\text{NiO}] = -240$; $\text{H}^\circ_{\text{at}}[\text{Ni}] = +430$; $(\text{IE1}+\text{IE2})[\text{Ni}] = +2490$; $\text{H}^\circ_{\text{at}}[\text{O}] = +249$; $(\text{EA1}+\text{EA2})[\text{O}] = +657$.

- (a)** Construct the cycle equation and calculate the lattice energy of NiO(s). **[4]**

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- (b)** Explain how the experimental lattice energy provides evidence for d-orbital crystal field stabilization energy in Ni^{2+} . **[2]**

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14 Iron(II) Sulfide Lattice Energy & Polarization — 9701/42/M/J/20/Q2 [6 Marks]

HARD · Syllabus 23.1

Iron(II) sulfide, FeS, has a nickel arsenide structure.
Data (kJ mol^{-1}): $\text{H}^\circ_{\text{f}} = -100$; $\text{H}^\circ_{\text{at}}[\text{Fe}] = +416$; $(\text{IE1}+\text{IE2})[\text{Fe}] = +2320$; $\text{H}^\circ_{\text{at}}[\text{S}] = +279$; $(\text{EA1}+\text{EA2})[\text{S}] = +332$.
Theoretical purely ionic LE = $-2780 \text{ kJ mol}^{-1}$.

- (a)** Calculate the experimental lattice energy of FeS(s). **[3]**

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- (b) Account for the substantial difference between the experimental and theoretical values. [3]

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15 Thermal Stability Trends of Group 2 Nitrates — 9701/41/M/J/20/Q2 [6 Marks]

HARD · Syllabus 23.4

Group 2 nitrates decompose: $2\text{M}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$.
Decomposition temperatures: $\text{Mg}(\text{NO}_3)_2 = 600 \text{ K}$; $\text{Ba}(\text{NO}_3)_2 = 900 \text{ K}$.

- (a) Explain why S° for the decomposition is positive and similar for all Group 2 nitrates. [2]

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- (b) Explain the trend in thermal stability down Group 2 in terms of cation polarising power. [4]

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16 Calcium Oxide Born–Haber Cycle & Hydration — 9701/42/O/N/19/Q2 [6 Marks]

HARD · Syllabus 23.1

Quicklime, CaO , reacts vigorously with water: $\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{s})$, $H^\circ = -65 \text{ kJ mol}^{-1}$.
Data (kJ mol^{-1}): $H^\circ_f[\text{CaO}] = -635$; $H^\circ_{\text{at}}[\text{Ca}] = +178$; $(\text{IE}_1 + \text{IE}_2)[\text{Ca}] = +1735$; $H^\circ_{\text{at}}[\text{O}] = +249$; $(\text{EA}_1 + \text{EA}_2)[\text{O}] = +657$.

- (a) Calculate the lattice energy of calcium oxide, $\text{CaO}(\text{s})$. [4]

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- (b) Explain why slaking of lime (adding water to CaO) is strongly exothermic. [2]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Lattice Energy: Sodium Chloride vs Magnesium Oxide — 9701/41/O/N/22/Q1 [4 Marks]

EASY · Syllabus 23.1

NaCl : $\text{LE} = -787 \text{ kJ mol}^{-1}$; MgO : $\text{LE} = -3791 \text{ kJ mol}^{-1}$. Both have rock-salt structures.

- (a) Explain why the lattice energy of MgO is almost five times more exothermic than that of NaCl . [4]

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18 Hydration Enthalpy of Bromide Ions in KBr — 9701/43/M/J/22/Q2 [4 Marks]

EASY · Syllabus 23.2

The dissolution of potassium bromide, KBr(s) , in water is represented by the enthalpy cycle in Fig. 18.1.
Data: $\text{LE}(\text{KBr}) = -679 \text{ kJ mol}^{-1}$; $\text{H}^\circ\text{sol}(\text{KBr}) = +20.0 \text{ kJ mol}^{-1}$; $\text{H}^\circ\text{hyd}(\text{K}^+) = -322 \text{ kJ mol}^{-1}$.

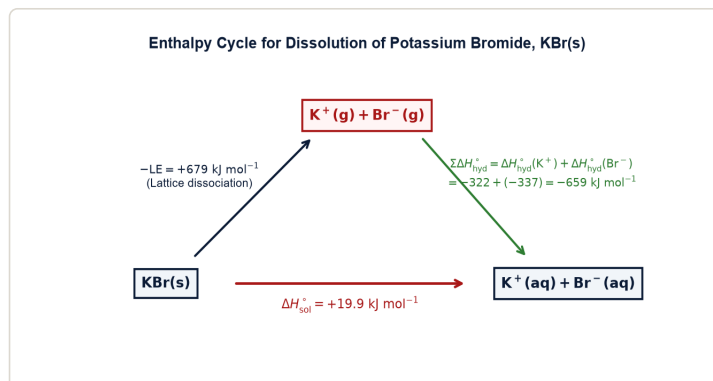


Fig. 18.1: Hess's Law enthalpy cycle relating lattice energy, ionic hydration, and enthalpy of solution for KBr(s) .

- (a) Calculate the standard enthalpy of hydration of the bromide ion, $\text{Br}^-(\text{g})$.

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19 Strontium Chloride Lattice Energy Calculation — 9701/43/O/N/21/Q1 [4 Marks]

EASY · Syllabus 23.1

Data (kJ mol^{-1}): $\text{H}^\circ\text{f}[\text{SrCl}_2] = -829$; $\text{H}^\circ\text{at}[\text{Sr}] = +164$; $(\text{IE1}+\text{IE2})[\text{Sr}] = +1614$; $\text{BE}(\text{Cl}_2) = +242$; $\text{EA}[\text{Cl}] = -349$.

- (a) Calculate the lattice energy of strontium chloride, $\text{SrCl}_2(\text{s})$.

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20 Group 2 Hydroxide Solubility Trend — 9701/42/M/J/21/Q1 [4 Marks]

EASY · Syllabus 23.2

Solubility of Group 2 hydroxides increases down Group 2 from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$.

- (a) Explain this trend in solubility in terms of the relative rates of decrease of LE and H°hyd .

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21 Second Electron Affinity of Oxygen from MgO Cycle — 9701/42/O/N/20/Q1 [4 Marks]

EASY · Syllabus 23.1

Data (kJ mol^{-1}): $\text{H}^\circ\text{f}[\text{MgO}] = -602$; $\text{H}^\circ\text{at}[\text{Mg}] = +148$; $(\text{IE1}+\text{IE2})[\text{Mg}] = +2187$; $\text{H}^\circ\text{at}[\text{O}] = +249$; $\text{EA1}[\text{O}] = -141$; $\text{LE}[\text{MgO}] = -3791$.

- (a) Calculate the value of the second electron affinity of oxygen, $EA_2[\text{O}]$. [4]

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22 Hydration Enthalpy: Magnesium vs Barium Ions — 9701/41/O/N/20/Q2 [4 Marks]

EASY · Syllabus 23.2

$H^\circ_{\text{hyd}}(\text{Mg}^{2+}) = -1920 \text{ kJ mol}^{-1}$ (radius 0.065 nm); $H^\circ_{\text{hyd}}(\text{Ba}^{2+}) = -1305 \text{ kJ mol}^{-1}$ (radius 0.135 nm).

- (a) Explain why H°_{hyd} of Mg^{2+} is significantly more exothermic than that of Ba^{2+} . [4]

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23 Entropy Change in Sodium Hydrogencarbonate Decomposition — 9701/42/M/J/20/Q1 [4 Marks]

EASY · Syllabus 23.3

$2\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$.
 $S^\circ (\text{J K}^{-1} \text{mol}^{-1})$: $\text{NaHCO}_3 = 102$; $\text{Na}_2\text{CO}_3 = 136$; $\text{H}_2\text{O}(\text{g}) = 189$; $\text{CO}_2(\text{g}) = 214$.

- (a) Calculate S° for this reaction and explain why it has a large positive value. [4]

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24 Barium Chloride Enthalpy of Formation — 9701/42/O/N/19/Q1 [4 Marks]

EASY · Syllabus 23.1

Data (kJ mol^{-1}): $H^\circ_{\text{at}}[\text{Ba}] = +180$; $(IE_1 + IE_2)[\text{Ba}] = +1468$; $BE(\text{Cl}_2) = +242$; $EA[\text{Cl}] = -349$; $LE[\text{BaCl}_2] = -2056$.

- (a) Calculate the standard enthalpy of formation, H°_{f} , of $\text{BaCl}_2(\text{s})$. [4]

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25 Copper(II) Sulfate Hydration Energy Cycle — 9701/41/O/N/19/Q2 [4 Marks]

EASY · Syllabus 23.2

$\text{CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\text{l}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$.
 $H^\circ_{\text{sol}}(\text{anhydrous}) = -66.5 \text{ kJ mol}^{-1}$; $H^\circ_{\text{sol}}(\text{hydrated}) = +11.7 \text{ kJ mol}^{-1}$.

- (a) Draw a Hess's Law cycle and calculate the standard enthalpy of hydration, H°_{hyd} . [4]

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26 Thermodynamics of Group 2 Sulfate Insolubility — 9701/41/O/N/18/Q2 [4 Marks]

EASY · Syllabus 23.2

MgSO_4 is highly soluble; BaSO_4 is insoluble.

(a)

Account for this solubility decrease down Group 2 by comparing changes in H°_{hyd} and lattice energy.

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27 Cesium Chloride vs Sodium Chloride Lattice Energy — 9701/42/M/J/17/Q1 [4 Marks]

EASY · Syllabus 23.1

NaCl : $\text{LE} = -787 \text{ kJ mol}^{-1}$; CsCl : $\text{LE} = -657 \text{ kJ mol}^{-1}$.

(a)

Explain why the lattice energy of CsCl is less exothermic than that of NaCl .

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28 Standard Entropy Change of Methanol Synthesis — 9701/41/M/J/17/Q2 [4 Marks]

EASY · Syllabus 23.3

$\text{CO(g)} + 2\text{H}_2\text{(g)} \rightarrow \text{CH}_3\text{OH(l)}$.

$S^\circ (\text{J K}^{-1} \text{mol}^{-1})$: $\text{CO(g)} = 197.6$; $\text{H}_2\text{(g)} = 130.6$; $\text{CH}_3\text{OH(l)} = 126.8$.

(a)

Calculate S° for this reaction and explain its negative sign.

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29 Rubidium Bromide Born–Haber Cycle Calculation — 9701/42/O/N/17/Q1 [4 Marks]

EASY · Syllabus 23.1

Data for RbBr (kJ mol^{-1}): $H^\circ_f = -395$; $H^\circ_{\text{at}}[\text{Rb}] = +81$; $IE_1[\text{Rb}] = +403$; $H^\circ_{\text{at}}[\frac{1}{2}\text{Br}_2] = +112$; $EA[\text{Br}] = -325$.

(a)

Calculate the lattice energy of rubidium bromide, RbBr(s) .

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30 Entropy Changes in Melting and Boiling of Water — 9701/41/O/N/17/Q2 [4 Marks]

EASY · Syllabus 23.3

For water: $S^\circ_{\text{fusion}} = +22.0 \text{ J K}^{-1} \text{mol}^{-1}$; $S^\circ_{\text{vaporization}} = +109.0 \text{ J K}^{-1} \text{mol}^{-1}$.

- (a) Explain why both entropy changes are positive, and why $S^\circ_{\text{vaporization}}$ is much larger than S°_{fusion} . [4]

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31 Magnesium Fluoride Lattice Energy Calculation — 9701/42/M/J/16/Q1 [4 Marks]

EASY · Syllabus 23.1

Data (kJ mol^{-1}): $H^\circ_f[\text{MgF}_2] = -1124$; $H^\circ_{\text{at}}[\text{Mg}] = +147$; $(IE_1 + IE_2)[\text{Mg}] = +2187$; $BE(\text{F}_2) = +158$; $EA[\text{F}] = -328$.

- (a) Calculate the lattice energy of magnesium fluoride, $\text{MgF}_2(\text{s})$. [4]

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32 Gibbs Free Energy of Iron Rusting — 9701/41/M/J/16/Q1 [4 Marks]

EASY · Syllabus 23.4

$4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$, $H^\circ = -1648 \text{ kJ mol}^{-1}$; $S^\circ = -544 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate G° at 298 K and state whether rusting is spontaneous at room temperature. [2]

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- (b) Calculate the temperature above which the reduction of Fe_2O_3 becomes feasible. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Definition of Enthalpy of Hydration — 9701/42/M/J/23/Q1(a) [2 Marks]

EASY · Syllabus 23.2

Hydration enthalpy governs solute-solvent energetics.

- (a) Define standard enthalpy change of hydration, H°_{hyd} , with an equation including state symbols for Mg^{2+} . [2]

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34 Definition and Sign of Lattice Energy — 9701/41/M/J/23/Q2(a) [2 Marks]

EASY · Syllabus 23.1

Lattice energy defines crystal lattice strength.

- (a) Define lattice energy and state its thermodynamic sign convention. [2]

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35 Entropy Change Sign in Vaporization — 9701/41/M/J/20/Q1(a) [2 Marks]

EASY · Syllabus 23.3

Entropy reflects microstate distribution.

- (a) State and explain the sign of S° for boiling water: $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$.

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36 Electron Affinities of Oxygen: Sign Distinction — 9701/42/O/N/18/Q1(a) [2 Marks]

EASY · Syllabus 23.1

Formation of oxide ions involves stepwise electron additions.

- (a) Explain why EA1 of oxygen is negative whereas EA2 is positive.

[2]

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37 Thermodynamic Criterion for Reaction Feasibility — 9701/41/M/J/19/Q1(a) [2 Marks]

EASY · Syllabus 23.4

Spontaneity is determined by the Second Law.

- (a) State the mathematical condition for feasibility in terms of G° , and state its SI units.

[2]

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38 Equilibrium Temperature Condition — 9701/43/O/N/22/Q1(a) [2 Marks]

EASY · Syllabus 23.4

At equilibrium, Gibbs free energy reaches a minimum.

- (a) Write the mathematical expression for the equilibrium temperature T in terms of H° and S° .

[2]

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39 Definition of Standard Molar Entropy — 9701/42/M/J/22/Q2(a) [2 Marks]

EASY · Syllabus 23.3

Third Law of Thermodynamics establishes absolute entropies.

- (a) Define standard molar entropy, S° , and explain why S° of an element is never zero at 298 K.

[2]

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40 Indirect Determination of Lattice Energy — 9701/41/O/N/21/Q1(a) [2 Marks]

EASY · Syllabus 23.1

Lattice energy cannot be measured in a simple calorimeter.

- (a) Explain why lattice energy cannot be measured directly by experiment.

[2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 HF 1: Barium Chloride Complete Born–Haber Cycle — 9701/42/O/N/23/Q1 [6 Marks]

HARD · Syllabus 23.1

HIGH-FREQUENCY CORE REPEAT #1 (Born–Haber Cycle Mastery)

The Born–Haber cycle for barium chloride, $\text{BaCl}_2(\text{s})$, is illustrated in Fig. 41.1.

Data for BaCl_2 (kJ mol^{-1}): $H^\circ_f = -859$; $H^\circ_{\text{at}}[\text{Ba}] = +180$; $IE_1[\text{Ba}] = +503$; $IE_2[\text{Ba}] = +965$; $BE(\text{Cl}_2) = +242$; $EA[\text{Cl}] = -349$.

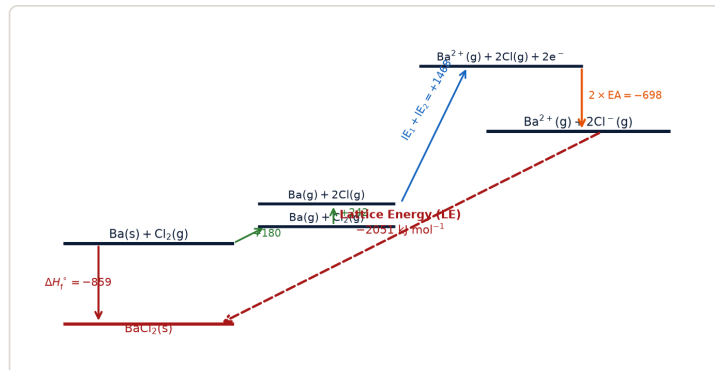


Fig. 41.1: Complete Born–Haber cycle energy level diagram for barium chloride, $\text{BaCl}_2(\text{s})$.

- (a) Construct the full Born–Haber cycle expression and calculate the lattice energy of $\text{BaCl}_2(\text{s})$.

[4]

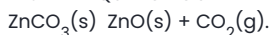
- (b) Compare the lattice energy of BaCl_2 with that of MgCl_2 ($LE = -2519 \text{ kJ mol}^{-1}$), explaining the difference.

[2]

42 HF 2: Thermal Decomposition & Feasibility of Zinc Carbonate — 9701/41/M/J/22/Q2 [6 Marks]

HARD · Syllabus 23.4

HIGH-FREQUENCY CORE REPEAT #2 (Gibbs Free Energy & Feasibility)



A plot of standard Gibbs free energy change G° against temperature for this decomposition is shown in Fig. 42.1.

Data: H°_f (kJ mol^{-1}): $\text{ZnCO}_3 = -813$; $\text{ZnO} = -348$; $\text{CO}_2 = -394$.

S° ($\text{J K}^{-1} \text{mol}^{-1}$): $\text{ZnCO}_3 = 82.4$; $\text{ZnO} = 43.6$; $\text{CO}_2 = 213.6$.

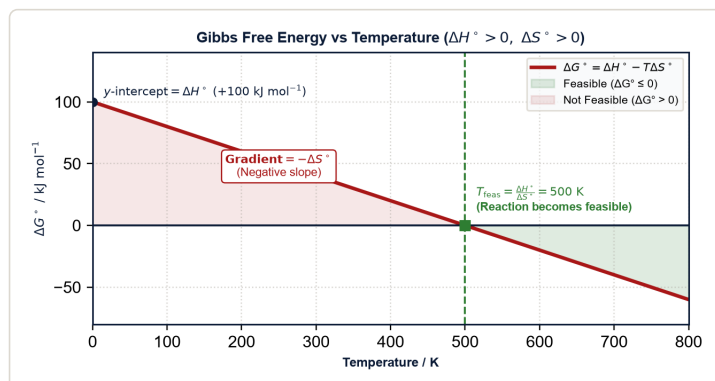


Fig. 42.1: Variation of standard Gibbs free energy change G° with temperature T showing the transition to feasibility.

- (a) Calculate the standard enthalpy change H° and standard entropy change S° for this reaction. [3]

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- (b) Calculate the minimum temperature, in $^\circ\text{C}$, at which the decomposition of ZnCO_3 becomes feasible. [3]

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43 HF 3: Enthalpy of Solution & Hydration of Sodium Fluoride — 9701/43/O/N/22/Q2 [6 Marks]

HARD · Syllabus 23.2

HIGH-FREQUENCY CORE REPEAT #3 (Enthalpy Cycle of Solution)

Lattice energy of $\text{NaF} = -918 \text{ kJ mol}^{-1}$.

Enthalpies of hydration: $H^\circ_{\text{hyd}}(\text{Na}^+) = -405 \text{ kJ mol}^{-1}$; $H^\circ_{\text{hyd}}(\text{F}^-) = -506 \text{ kJ mol}^{-1}$.

- (a) Construct an energy cycle and calculate the standard enthalpy of solution, H°_{sol} , of NaF(s) . [3]

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- (b) Explain why NaF is significantly less soluble than NaCl ($\text{LE} = -787 \text{ kJ mol}^{-1}$; $H^\circ_{\text{sol}} = +4.0 \text{ kJ mol}^{-1}$). [3]

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44 HF 4: Comprehensive Group 2 Sulfate Solubility Analysis — 9701/42/M/J/21/Q2 [6 Marks]

HARD · Syllabus 23.2

HIGH-FREQUENCY CORE REPEAT #4 (Sulfate Solubility Rationale)

The solubility of Group 2 sulfates decreases down Group 2 from MgSO_4 (soluble) to BaSO_4 (insoluble). Fig. 44.1 shows how the magnitudes of hydration enthalpy and lattice energy change with increasing cation radius down the group.

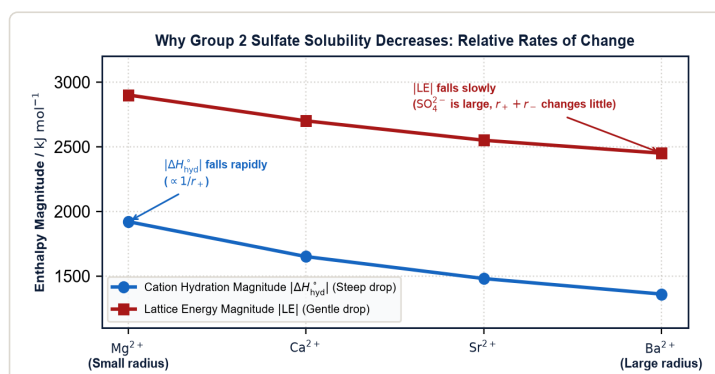


Fig. 44.1: Comparative rates of decline in $|H^\circ_{\text{hyd}}|$ and $|\text{LE}|$ down Group 2 explaining sulfate insolubility.

- (a) Define standard enthalpy change of solution, H°_{sol} . [2]

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- (b) Explain in detail, using enthalpy cycles and ionic radii, why the solubility of sulfates decreases down the group.

[4]

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45 HF 5: Theoretical vs Experimental Lattice Energy in Silver Halides — 9701/42/O/N/21/Q2 [4 Marks]

EASY · Syllabus 23.1

HIGH-FREQUENCY CORE REPEAT #5 (Polarization in Silver Halides)

For AgI: Experimental LE = -889 kJ mol^{-1} ; Theoretical LE = -736 kJ mol^{-1} .

For AgF: Experimental LE = -958 kJ mol^{-1} ; Theoretical LE = -920 kJ mol^{-1} .

- (a) Explain why the discrepancy between experimental and theoretical lattice energy is much greater for AgI than for AgF.

[4]

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46 HF 6: Endothermic Nature of Second Electron Affinity — 9701/41/M/J/21/Q2 [4 Marks]

EASY · Syllabus 23.1

HIGH-FREQUENCY CORE REPEAT #6 (Electron Affinities of Group 16)

For sulfur: $\text{S(g)} + \text{e}^- \rightarrow \text{S}^-(\text{g})$, EA1 = -200 kJ mol^{-1} ;

$\text{S}^-(\text{g}) + \text{e}^- \rightarrow \text{S}^{2-}(\text{g})$, EA2 = $+532 \text{ kJ mol}^{-1}$.

- (a) Explain why the first electron affinity of sulfur is exothermic while the second electron affinity is endothermic.

[4]

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47 HF 7: Entropy Change in Carbonate Acidification — 9701/43/M/J/20/Q1 [4 Marks]

EASY · Syllabus 23.3

HIGH-FREQUENCY CORE REPEAT #7 (Entropy of Gas Evolution)

$\text{MgCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$.

- (a) Predict and explain the sign of S° for this reaction, and explain how an increase in temperature affects the spontaneity.

[4]

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48 HF 8: Group 2 Cation Hydration Enthalpy Trends — 9701/42/O/N/19/Q1 [4 Marks]

EASY · Syllabus 23.2

HIGH-FREQUENCY CORE REPEAT #8 (Cation Charge Density & Hydration)

Values of H°_{hyd} : $\text{Be}^{2+} = -2494$; $\text{Mg}^{2+} = -1920$; $\text{Ca}^{2+} = -1577$; $\text{Sr}^{2+} = -1443$; $\text{Ba}^{2+} = -1305 \text{ kJ mol}^{-1}$.

- (a) Explain why H°_{hyd} becomes progressively less exothermic down Group 2 from Be^{2+} to Ba^{2+} . [4]

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49 HF 9: Standard Definition of Lattice Energy — 9701/41/M/J/23/Q1(a) [2 Marks]

EASY · Syllabus 23.1

HIGH-FREQUENCY CORE REPEAT #9 (Core Definition)

- (a) Define lattice energy with an equation including state symbols for calcium oxide, CaO . [2]

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50 HF 10: Thermodynamic Feasibility & Spontaneity Condition — 9701/42/M/J/22/Q1(a) [2 Marks]

EASY · Syllabus 23.4

HIGH-FREQUENCY CORE REPEAT #10 (Feasibility Criterion)

- (a) State the thermodynamic condition for a reaction to be feasible at temperature T in terms of Gibbs free energy, and explain why some feasible reactions do not occur at observable rates. [2]

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Mark Scheme — Topic 23 — Chemical Energetics

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Enthalpy change when 1 mole of gaseous ions [1] is completely hydrated in water to form infinitely dilute aqueous solution under standard conditions [1].	2
1(b)	$H^{\circ}f = H^{\circ}at(Zn) + (IE1+IE2)(Zn) + H^{\circ}at(S) + (EA1+EA2)(S) + LE(ZnS)$ [1]; $-206 = +131 + 2640 + 279 + 332 + LE$ [1]; $LE = -206 - 3382$ [1]; $LE(ZnS) = -3588 \text{ kJ mol}^{-1}$ [1].	4
2(a)	$H^{\circ}f = H^{\circ}at(Mg) + (IE1+IE2) + 2(H^{\circ}at(Cl)) + 2(EA(Cl)) + LE$ [1]; $-641 = 147 + 2187 + 242 + 2(-349) + LE = 1878 + LE$ [1]; $LE(MgCl_2) = -2519 \text{ kJ mol}^{-1}$ [1].	3
2(b)	$S^{\circ} = 32.7 + 223.0 - 90.0 = +165.7 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]; At feasibility, $T = H^{\circ} / S^{\circ}$ [1]; $T = (641 \times 1000) / 165.7 = 3868 \text{ K}$ [1].	3
3(a)	$-1220 = 178 + 1735 + 158 + 2(-328) + LE = 1415 + LE$ [1]; $LE(CaF_2) = -2635 \text{ kJ mol}^{-1}$ [2].	3
3(b)	$H^{\circ}sol = [-1577 + 2(-506)] - (-2635) = -2589 + 2635 = +46 \text{ kJ mol}^{-1}$ [1]; $H^{\circ}sol$ is endothermic [1]; F^- is very small, giving CaF_2 a vastly more exothermic LE that outweighs hydration enthalpy [1].	3
4(a)	$H^{\circ}sol = H^{\circ}hyd(Ba^{2+}) + H^{\circ}hyd(SO_4^{2-}) - LE$ [1]; $H^{\circ}sol = (-1305 - 1145) - (-2469) = -2450 + 2469$ [1]; $H^{\circ}sol = +19 \text{ kJ mol}^{-1}$ [1].	3
4(b)	Both LE and $H^{\circ}hyd$ become less exothermic down the group as cation radius increases [1]; however, $H^{\circ}hyd$ decreases more rapidly than LE because SO_4^{2-} is very large [1]; hence $H^{\circ}sol$ becomes more endothermic down the group, decreasing solubility [1].	3
5(a)	$-127 = 285 + 731 + 121 + (-349) + LE = 788 + LE$ [1]; $LE(AgCl) = -915 \text{ kJ mol}^{-1}$ [2].	3
5(b)	Experimental LE (-915) is significantly more exothermic than theoretical LE (-770) [1]; Ag^+ has high polarising power due to poorly shielding 4d10 electrons, polarising the Cl^- ion [1]; introduces significant covalent character, strengthening the lattice [1].	3
6(a)	$-169 = 2(338) + 2(746) + 249 + (-141 + 798) + LE = 676 + 1492 + 249 + 657 + LE$ [2]; $-169 = 3074 + LE$ & $LE = -3243 \text{ kJ mol}^{-1}$ [2].	4
6(b)	$EA1$ is exothermic due to attraction between incoming electron and positive nucleus [1]; $EA2$ is endothermic because adding an electron to negative O^- requires energy to overcome electrostatic repulsion [1].	2
7(a)	$S^{\circ} = (113 + 146) - 151 = 259 - 151 = +108 \text{ J K}^{-1} \text{ mol}^{-1}$ [3].	3
7(b)	$G^{\circ} = +25.7 - (298 \times 0.108) = 25.7 - 32.18 = -6.48 \text{ kJ mol}^{-1}$ [2]; Spontaneous because G° is negative; the TS° term outweighs the endothermic H° term (entropy driven) [1].	3
8(a)	$H^{\circ} = [-635 + (-394)] - (-1207) = +178 \text{ kJ mol}^{-1}$ [1]; $S^{\circ} = (39.7 + 213.6) - 92.9 = +160.4 \text{ J K}^{-1} \text{ mol}^{-1}$ [2].	3
8(b)	$T = H^{\circ} / S^{\circ} = (178 \times 1000) / 160.4 = 1109.7 \text{ K}$ [2]; $T(^{\circ}C) = 1109.7 - 273 = 837^{\circ}C$ [1].	3
9(a)	$-270 = 161 + 520 + 107 + (-295) + LE = 493 + LE$ [1]; $LE(LiI) = -763 \text{ kJ mol}^{-1}$ [2].	3
9(b)	Experimental value (-763) is more exothermic than theoretical (-738) [1]; Li^+ has very small radius and high charge density (high polarising power) [1]; I^- has large polarisable electron cloud, distorted by Li^+ to give covalent character [1].	3
10(a)	$-1676 = 2(326) + 2(5140) + 1.5(496) + 3(657) + LE = 13647 + LE$ [2]; $LE(Al_2O_3) = -15323 \text{ kJ mol}^{-1}$ [2].	4
10(b)	Al^{3+} has a +3 charge and smaller radius than Na^+ (+1), and O^{2-} has a -2 charge [1]; electrostatic attraction is proportional to charge product (+6 vs +2) and inversely proportional to ionic radii sum [1].	2
11(a)	$S^{\circ} = 2(192.3) - [191.6 + 3(130.6)] = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ [2].	2
11(b)	$G^{\circ} = -92.2 - [298 \times (-0.1988)] = -92.2 + 59.24 = -32.96 \text{ kJ mol}^{-1}$ [1]; Spontaneous because $G^{\circ} < 0$ [1].	2
11(c)	$T = H^{\circ} / S^{\circ} = (-92.2 \times 1000) / (-198.8) = 463.8 \text{ K}$ ($190.8^{\circ}C$) [2].	2
12(a)	$H^{\circ}sol = [-322 + (-364)] - (-711) = -686 + 711 = +25.0 \text{ kJ mol}^{-1}$ [2].	2
12(b)	$S^{\circ}sol = (102.5 + 56.5) - 82.6 = 159.0 - 82.6 = +76.4 \text{ J K}^{-1} \text{ mol}^{-1}$ [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
12(c)	$G^\circ = +25.0 - (298 \times 0.0764) = 25.0 - 22.77 = +2.23 \text{ kJ mol}^{-1}$ [1]; Dissolves readily because G is close to zero and non-standard entropy of dilution drives dissolution to saturation [1].	2
13(a)	$-240 = 430 + 2490 + 249 + 657 + LE = 3826 + LE$ [2]; $LE(\text{NiO}) = -240 - 3826 = -4066 \text{ kJ mol}^{-1}$ [2].	4
13(b)	Experimental LE is more exothermic than predicted by simple ionic radius extrapolation across Period 4 [1]; extra stability arises from crystal field splitting of d-orbitals in octahedral oxide environment (CFSE of d8 ion) [1].	2
14(a)	$-100 = 416 + 2320 + 279 + 332 + LE = 3347 + LE$ [1]; $LE(\text{FeS}) = -3447 \text{ kJ mol}^{-1}$ [2].	3
14(b)	Experimental LE (-3447) is much more exothermic than theoretical LE (-2780) [1]; Fe^{2+} cation has high polarising power due to incomplete d-subshell [1]; polarises large, soft S^{2-} ion to introduce major covalent character [1].	3
15(a)	Decomposition produces 5 moles of gas ($4\text{NO}_2 + \text{O}_2$) from 2 moles of solid, creating large dispersal of matter/microstates [1]; stoichiometry is identical for all Group 2 nitrates [1].	2
15(b)	Descending Group 2, cation radius increases while ionic charge remains +2 [1]; charge density and polarising power of M^{2+} decrease down the group [1]; Mg^{2+} distorts the electron cloud of the NO_3^- anion more strongly than Ba^{2+} [1]; weakens N–O bond within nitrate, requiring lower thermal energy to decompose [1].	4
16(a)	$-635 = 178 + 1735 + 249 + 657 + LE = 2819 + LE$ [2]; $LE(\text{CaO}) = -635 - 2819 = -3454 \text{ kJ mol}^{-1}$ [2].	4
16(b)	O^{2-} reacts with water to form OH^- ions with release of strong acid-base proton transfer energy [1]; accompanied by highly exothermic hydration of Ca^{2+} and OH^- ions into the $\text{Ca}(\text{OH})_2$ crystal [1].	2
17(a)	Lattice energy is proportional to $(q^+ \times q^-) / (r^+ + r^-)$ [1]; Mg^{2+} and O^{2-} have +2 and -2 charges (product = 4), whereas Na^+ and Cl^- have +1 and -1 (product = 1) [1]; Mg^{2+} (0.065 nm) is smaller than Na^+ (0.095 nm), and O^{2-} (0.140 nm) is smaller than Cl^- (0.181 nm) [1]; smaller inter-ionic distance and quadrupled charge product produce vastly stronger electrostatic attraction [1].	4
18(a)	$H^\circ_{\text{sol}} = H^\circ_{\text{hyd}}(\text{K}^+) + H^\circ_{\text{hyd}}(\text{Br}^-) - LE$ [1]; $+20.0 = -322 + H^\circ_{\text{hyd}}(\text{Br}^-) - (-679)$ [1]; $+20.0 = +357 + H^\circ_{\text{hyd}}(\text{Br}^-)$ [1]; $H^\circ_{\text{hyd}}(\text{Br}^-) = -337 \text{ kJ mol}^{-1}$ [1].	4
19(a)	$H^\circ_{\text{f}} = H^\circ_{\text{at}}(\text{Sr}) + (IE_1 + IE_2) + BE(\text{Cl}_2) + 2 \times EA(\text{Cl}) + LE$ [1]; $-829 = 164 + 1614 + 242 + 2(-349) + LE$ [1]; $-829 = 1322 + LE$ [1]; $LE(\text{SrCl}_2) = -2151 \text{ kJ mol}^{-1}$ [1].	4
20(a)	Cation radius increases down Group 2, so both LE and H°_{hyd} become less exothermic [1]; OH^- is a relatively small anion, so $(r^+ + r^-)$ increases significantly [1]; LE decreases more rapidly than H°_{hyd} down the group [1]; $H^\circ_{\text{sol}} = H^\circ_{\text{hyd}} - LE$ becomes more exothermic / less endothermic, increasing solubility [1].	4
21(a)	$-602 = 148 + 2187 + 249 + (-141) + EA_2 + (-3791)$ [1]; $-602 = -1348 + EA_2$ [2]; $EA_2 = +746 \text{ kJ mol}^{-1}$ [1].	4
22(a)	Hydration involves ion-dipole attractions between the cation and water molecules [1]; both cations have a +2 charge [1]; Mg^{2+} has a much smaller ionic radius and higher charge density [1]; creates stronger ion-dipole attractions, releasing more energy upon hydration [1].	4
23(a)	$S^\circ = [136 + 189 + 214] - 2(102) = 539 - 204 = +335 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]; Reaction converts 2 moles of solid into 1 mole of solid and 2 moles of gas [1]; gases possess vastly higher disorder and translational microstates than crystalline solids [1].	4
24(a)	$H^\circ_{\text{f}} = 180 + 1468 + 242 + 2(-349) + (-2056)$ [2]; $H^\circ_{\text{f}} = 1890 - 698 - 2056$ [1]; $H^\circ_{\text{f}} = -864 \text{ kJ mol}^{-1}$ [1].	4
25(a)	Hess's Law cycle connecting both solids to aqueous solution [1]; $H^\circ_{\text{hyd}} = H^\circ_{\text{sol}}(\text{anhydrous}) - H^\circ_{\text{sol}}(\text{hydrated})$ [1]; $H^\circ_{\text{hyd}} = -66.5 - (+11.7)$ [1]; $H^\circ_{\text{hyd}} = -78.2 \text{ kJ mol}^{-1}$ [1].	4
26(a)	$H^\circ_{\text{sol}} = H^\circ_{\text{hyd}} - LE$ [1]; descending group, cation radius increases, so both LE and H°_{hyd} become less exothermic [1]; because SO_4^{2-} is very large, LE decreases very slowly [1]; H°_{hyd} drops rapidly with increasing cation radius, making H°_{sol} significantly more endothermic, reducing solubility [1].	4
27(a)	Both salts contain ions with +1 and -1 charges [1]; Cs^+ has a significantly larger ionic radius (0.167 nm) than Na^+ (0.095 nm) [1]; inter-ionic distance ($r^+ + r^-$) is greater in CsCl [1]; electrostatic force of attraction between ions is weaker, releasing less energy when crystal lattice forms [1].	4
28(a)	$S^\circ = 126.8 - [197.6 + 2(130.6)] = 126.8 - 458.8$ [2]; $S^\circ = -332.0 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]; Negative because 3 moles of gas react to form 1 mole of liquid, causing an immense decrease in disorder/microstates [1].	4
29(a)	$H^\circ_{\text{f}} = 81 + 403 + 112 + (-325) + LE = 271 + LE$ [2]; $-395 = 271 + LE$ [1]; $LE(\text{RbBr}) = -666 \text{ kJ mol}^{-1}$ [1].	4

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
30(a)	Both involve increase in disorder as molecules gain rotational/translational freedom [1]; fusion involves moving from fixed lattice to liquid where hydrogen bonds largely persist [1]; vaporization completely breaks intermolecular hydrogen bonds [1]; gas molecules possess complete translational randomness, yielding an immense increase in microstates [1].	4
31(a)	$-1124 = 147 + 2187 + 158 + 2(-328) + LE = 1836 + LE$ [2]; $LE(MgF_2) = -1124 - 1836$ [1]; $LE = -2960 \text{ kJ mol}^{-1}$ [1].	4
32(a)	$G^\circ = -1648 - [298 \times (-0.544)] = -1648 + 162.1 = -1485.9 \text{ kJ mol}^{-1}$ [1]; Spontaneous because G° is highly negative [1].	2
32(b)	$T = H^\circ / S^\circ = (-1648 \times 1000) / (-544) = 3029 \text{ K (2756 }^\circ\text{C)}$ [2].	2
33(a)	Enthalpy change when 1 mole of gaseous ions dissolves in water to form infinitely dilute solution under standard conditions [1]; $Mg^{2+}(g) + aq \rightarrow Mg^{2+}(aq)$ [1].	2
34(a)	Enthalpy change when 1 mole of an ionic crystalline solid is formed from its constituent gaseous ions under standard conditions [1]; always negative (exothermic) [1].	2
35(a)	$S^\circ > 0$ (positive) [1]; gas phase has complete translational freedom and vastly more microstates than liquid [1].	2
36(a)	EA1 is exothermic due to nuclear attraction [1]; EA2 is endothermic due to electrostatic repulsion between incoming electron and O^- ion [1].	2
37(a)	$G^\circ \leq 0$ (negative or zero) [1]; units: kJ mol^{-1} (or J mol^{-1}) [1].	2
38(a)	At equilibrium $G^\circ = 0$ & Arr; $H^\circ - TS^\circ = 0$ [1]; $T = H^\circ / S^\circ$ (with H in J and S in J K^{-1}) [1].	2
39(a)	The entropy of 1 mole of a substance in its standard state at 298 K and 100 kPa [1]; at 298 K atoms possess thermal vibrational/translational energy and disorder ($S^\circ = 0$ only at 0 K for perfect crystal) [1].	2
40(a)	It is impossible to isolate and condense separate gaseous ions directly into a solid crystal lattice without other competing processes [1]; ions immediately recombine or discharge [1].	2
41(a)	$H^\circ_f = H^\circ_{at}(Ba) + (IE_1 + IE_2)(Ba) + BE(Cl_2) + 2 \times EA(Cl) + LE$ [1]; $-859 = 180 + 1468 + 242 + 2(-349) + LE = 1192 + LE$ [1]; $LE(BaCl_2) = -859 - 1192$ [1]; $LE = -2051 \text{ kJ mol}^{-1}$ [1].	4
41(b)	LE of $BaCl_2$ is less exothermic than $MgCl_2$ [1]; Ba^{2+} has a larger ionic radius than Mg^{2+} , resulting in greater inter-ionic separation and weaker electrostatic attraction [1].	2
42(a)	$H^\circ = [-348 + (-394)] - (-813) = -742 + 813 = +71.0 \text{ kJ mol}^{-1}$ [1]; $S^\circ = (43.6 + 213.6) - 82.4 = 257.2 - 82.4 = +174.8 \text{ J K}^{-1} \text{ mol}^{-1}$ [2].	3
42(b)	$T = H^\circ / S^\circ = (71.0 \times 1000) / 174.8 = 406.2 \text{ K}$ [2]; $T(^{\circ}\text{C}) = 406.2 - 273 = 133.2 ^{\circ}\text{C}$ (or $133 ^{\circ}\text{C}$) [1].	3
43(a)	$H^\circ_{sol} = H^\circ_{hyd}(Na^+) + H^\circ_{hyd}(F^-) - LE$ [1]; $H^\circ_{sol} = (-405 - 506) - (-918) = -911 + 918$ [1]; $H^\circ_{sol} = +7.0 \text{ kJ mol}^{-1}$ [1].	3
43(b)	F^- is much smaller than Cl^- , leading to a vastly more exothermic lattice energy in NaF (-918 vs -787) [1]; this large LE requirement makes dissolving NaF less thermodynamically favourable [1]; hydration enthalpy cannot fully compensate for the high lattice stability [1].	3
44(a)	Enthalpy change when 1 mole of an ionic compound dissolves in water to form infinitely dilute solution under standard conditions [2].	2
44(b)	$H^\circ_{sol} = H^\circ_{hyd} - LE$ [1]; down Group 2, cation radius increases (Mg^{2+} to Ba^{2+}), so both LE and H°_{hyd} become less exothermic [1]; sulfate SO_4^{2-} is very large, so cation size increase causes only small percentage change in $(r_+ + r_-)$, meaning LE decreases slowly [1]; cation H°_{hyd} depends sharply on $1/r_+$ and drops rapidly; hence H°_{sol} becomes more endothermic, reducing solubility [1].	4
45(a)	Theoretical model assumes purely spherical, non-polarisable ionic bonding [1]; Ag^+ cation has high polarising power due to poorly shielding 4d10 electrons [1]; I^- has a vastly larger, more polarisable electron cloud than F^- [1]; electron cloud of I^- is severely distorted towards Ag^+ , introducing substantial covalent character that reinforces the lattice [1].	4
46(a)	EA1 involves an electron added to a neutral gaseous atom [1]; energy is released due to the electrostatic attraction between the incoming electron and positive nucleus [1]; EA2 involves adding an electron to a negatively charged ion (S^-) [1]; significant electrostatic repulsion between the negative ion and incoming electron must be overcome, requiring net energy input [1].	4

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
47(a)	S° is positive [1]; gas (CO_2) is produced from solid and solution, substantially increasing the number of microstates and dispersal of matter [1]; since S° is positive, $-TS^\circ$ becomes increasingly negative as temperature rises [1]; this makes $G^\circ = H^\circ - TS^\circ$ more negative, enhancing thermodynamic spontaneity [1].	4
48(a)	Hydration enthalpy depends on electrostatic attraction between the metal cation and water dipoles [1]; all Group 2 ions have identical +2 charge [1]; ionic radius increases down the group (Be^{2+} to Ba^{2+}), so charge density decreases [1]; weaker electrostatic attraction between the larger cation and the partial negative oxygen atom of water releases less energy [1].	4
49(a)	The enthalpy change when 1 mole of an ionic crystalline solid is formed from its constituent gaseous ions under standard conditions [1]; $\text{Ca}^{2+}(\text{g}) + \text{O}^{2-}(\text{g}) \rightarrow \text{CaO}(\text{s})$ [1].	2
50(a)	$G^\circ \leq 0$ (Gibbs free energy change must be negative or zero) [1]; the reaction may have a very high activation energy (kinetic stability) preventing reaction at room temperature [1].	2